

Electromagnetic Waves

Problem 9.1

By explicit differentiation, check that the functions f_1, f_2 , and f_3 satisfy the wave equation. Show that f_4 and f_5 do not.

$$f_1(z, t) = Ae^{-b(z-vt)^2}$$

$$f_2(z, t) = A \sin[b(z - vt)]$$

$$f_3(z, t) = \frac{A}{b(z - vt)^2 + 1}$$

$$f_4(z, t) = Ae^{-b(bz^2 + vt)}$$

$$f_5(z, t) = A \sin(bz) \cos(bvt)^3$$

Solution:

The wave equation is

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}.$$

f_1 :

$$\begin{aligned} \frac{\partial f_1}{\partial z} &= -2bA(z - vt)e^{-b(z-vt)^2}, & \frac{\partial^2 f_1}{\partial z^2} &= 4b^2A(z - vt)^2e^{-b(z-vt)^2} \\ \frac{\partial f_1}{\partial t} &= 2vbA(z - vt)e^{-b(z-vt)^2}, & \frac{\partial^2 f_1}{\partial t^2} &= 4v^2b^2A(z - vt)^2e^{-b(z-vt)^2} \end{aligned}$$

$$\frac{\partial^2 f_1}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f_1}{\partial t^2}$$

↓

$$4b^2A(z - vt)^2e^{-b(z-vt)^2} = \frac{1}{v^2} 4v^2b^2A(z - vt)^2e^{-b(z-vt)^2} \checkmark$$

f_2 :

$$\begin{aligned}\frac{\partial f_2}{\partial z} &= bA \cos[b(z - vt)], & \frac{\partial^2 f_2}{\partial z^2} &= -b^2 A \sin[b(z - vt)] \\ \frac{\partial f_2}{\partial t} &= -vbA \cos[b(z - vt)], & \frac{\partial^2 f_2}{\partial t^2} &= -v^2 b^2 A \sin[b(z - vt)]\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f_2}{\partial z^2} &= \frac{1}{v^2} \frac{\partial^2 f_2}{\partial t^2} \\ \downarrow \\ -b^2 A \sin[b(z - vt)] &= -\frac{1}{v^2} v^2 b^2 A \sin[b(z - vt)] \checkmark\end{aligned}$$

f_3 using quotient rule:

$$\begin{aligned}\frac{\partial f_3}{\partial z} &= \frac{-2bA(z - vt)}{(b(z - vt)^2 + 1)^2} \\ \frac{\partial^2 f_3}{\partial z^2} &= \frac{-2bA}{(b(z - vt)^2 + 1)^2} - \frac{-2bA(z - vt)2(b(z - vt)^2 + 1)(2b(z - vt))}{(b(z - vt)^2 + 1)^4} = \\ &= \frac{-2bA}{(b(z - vt)^2 + 1)^2} + \frac{8b^2 A(z - vt)^2}{(b(z - vt)^2 + 1)^3} \\ \frac{\partial f_3}{\partial t} &= \frac{2bvA(z - vt)}{(b(z - vt)^2 + 1)^2} \\ \frac{\partial^2 f_3}{\partial t^2} &= \frac{-2bv^2 A}{(b(z - vt)^2 + 1)^2} - \frac{2bvA(z - vt)2(b(z - vt)^2 + 1)(-2bv(z - vt))}{(b(z - vt)^2 + 1)^4} = \\ &= \frac{-2bv^2 A}{(b(z - vt)^2 + 1)^2} + \frac{8b^2 v^2 A(z - vt)^2}{(b(z - vt)^2 + 1)^3}\end{aligned}$$

$$\frac{\partial^2 f_3}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f_3}{\partial t^2} \checkmark$$

f_4 :

$$\begin{aligned}\frac{\partial f_4}{\partial z} &= -2b^2 z A e^{-b(bz^2 + vt)}, & \frac{\partial^2 f_4}{\partial z^2} &= (-2b^2 + 4b^4 z^2) (A e^{-b(bz^2 + vt)}) \\ \frac{\partial f_4}{\partial t} &= -bvA e^{-b(bz^2 + vt)}, & \frac{\partial^2 f_4}{\partial t^2} &= b^2 v^2 A e^{-b(bz^2 + vt)}\end{aligned}$$

$$\frac{\partial^2 f_4}{\partial z^2} \neq \frac{1}{v^2} \frac{\partial^2 f_4}{\partial t^2} \times$$

f_5 :

$$\begin{aligned}\frac{\partial f_5}{\partial z} &= bA \cos(bz) \cos(bvt)^3, & \frac{\partial^2 f_5}{\partial z^2} &= -b^2 A \sin(bz) \cos(bvt)^3 \\ \frac{\partial f_5}{\partial t} &= -3(b^3 v^3 t^2) \sin(bz) \sin(bvt)^3 \\ \frac{\partial^2 f_5}{\partial t^2} &= -9(b^6 v^6 t^4) \sin(bz) \cos(bvt)^3 - 6(b^3 v^3 t) \sin(bz) \sin(bvt)^3\end{aligned}$$

$$\frac{\partial^2 f_5}{\partial z^2} \neq \frac{1}{v^2} \frac{\partial^2 f_5}{\partial t^2} \times$$

□

Problem 9.2

Show that the **standing wave** $f(z, t) = A \sin(kz) \cos(kvt)$ satisfies the wave equation, and express it as the sum of a wave traveling to the left and a wave traveling to the right.

Solution:

$$\begin{aligned}\frac{\partial^2 f}{\partial z^2} &= -k^2 A \sin(kz) \cos(kvt) \\ \frac{\partial^2 f}{\partial t^2} &= -v^2 k^2 A \sin(kz) \cos(kvt)\end{aligned}$$

The wave equation

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}$$

is satisfied. Using the trigonometric identity

$$\sin(a) \cos(b) = \frac{1}{2} [\sin(a+b) + \sin(a-b)]$$

we obtain

$$\frac{A}{2} [\sin(k(z+vt)) + \sin(k(z-vt))]$$

which is the sum of a wave traveling to the right and left. □

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